# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 6 & 10: **Triangles, Circles** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** A 1.5 m boy casts a 2 m shadow. At the same time a pole casts a 10 m shadow. The height of the pole is: 1

**(a)** 5 m **(b)** 13.3 m **(c)** 7.5 m **(d)** 6 m

**2.** If $\triangle ABC \sim \triangle DEF$ with $\angle B = 70^\circ$ and $\angle C = 50^\circ$, then $\angle D$ is: 1

**(a)** $50^\circ$ **(b)** $70^\circ$ **(c)** $60^\circ$ **(d)** $120^\circ$

**3.** In $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{AB} = \frac{3}{5}$ and $DE = 6$ cm. Then BC is: 1

**(a)** 3.6 cm **(b)** 10 cm **(c)** 9 cm **(d)** 15 cm

**4.** A student states: "If two angles of one triangle equal two angles of another, the triangles are congruent." The correct response is: 1

**(a)** Correct
**(b)** Correct for equilateral triangles
**(c)** Incorrect — they are similar, not necessarily congruent
**(d)** Cannot be determined

**5.** Student A says all equilateral triangles are similar; Student B says only if their sides are equal. Then: 1

**(a)** Student A is correct **(b)** Student B is correct
**(c)** Both are correct **(d)** Both are wrong

**6.** The maximum number of common points of a tangent and a circle is: 1

**(a)** infinitely many **(b)** 0 **(c)** 1 **(d)** 2

**7.** PA and PB are tangents to a circle of radius 6 cm with $\angle APB = 90^\circ$. Then OP is: 1

**(a)** $3\sqrt{2}$ cm **(b)** 6 cm **(c)** 12 cm **(d)** $6\sqrt{2}$ cm

**8.** The length of the tangent from a point 26 cm from the centre of a circle of radius 10 cm is: 1

**(a)** 24 cm **(b)** 36 cm **(c)** 20 cm **(d)** 16 cm

**9.** A student states: "A secant can touch a circle at exactly one point." The correct response is: 1

**(a)** Incorrect — a secant cuts the circle at two points
**(b)** Correct for small circles
**(c)** Correct
**(d)** Cannot be determined

**10.** Tangents TP and TQ make $\angle PTQ = 60^\circ$. For $\angle OPQ$, Student A says $30^\circ$; Student B says $60^\circ$. Then: 1

**(a)** Student A is correct **(b)** Student B is correct
**(c)** Both are correct **(d)** Both are wrong

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** In $\triangle ABC$, $DE \parallel BC$ with $AD = 6$ cm, $DB = 9$ cm and $AE = 8$ cm. Find AC. **2**

**12.** $\triangle ABC$ has $\angle A = 60^\circ, \angle B = 80^\circ$ and $\triangle PQR$ has $\angle Q = 80^\circ, \angle R = 40^\circ$. Are they similar? Write the correspondence. **2**

**13.** The tangent from a point 25 cm from the centre of a circle is 24 cm long. Find the radius. **2**

**14.** TP and TQ are tangents to a circle with centre O and $\angle POQ = 110^\circ$. Find $\angle TPQ$. **2**

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Diagonal BD of parallelogram ABCD intersects AE at F, where E is a point on BC. Prove that $DF \times EF = FB \times FA$. **3**

**16.** Prove that the tangents drawn from an external point to a circle subtend equal angles at the centre. **3**

**17.** A vertical pole 10 m high casts an 8 m shadow when a tower casts a 40 m shadow. Find the height of the tower. What will be the tower's shadow when the pole's shadow is 10 m? **3**

**OR**

In $\triangle ABC$, P and Q are on AB and AC with $AP = 3$, $PB = 6$, $AQ = 5$ and $QC = 10$ cm. Show that $PQ \parallel BC$ and find PQ if $BC = 12$ cm. **3**

**18.** A quadrilateral ABCD circumscribes a circle with $AB = 2x$, $BC = x + 4$, $CD = 9$ and $DA = 7$ cm. Find $x$ and the sides AB and BC. **3**

**OR**

Prove that the line segment joining the points of contact of two parallel tangents of a circle passes through its centre. **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study:** To find the width of a river, A is a point on the near side; P and Q are on the near bank, and B and C on the far bank, with A, P, B collinear, A, Q, C collinear and $PQ \parallel BC$. $AP = 20$ m, $PQ = 30$ m and $BC = 75$ m. **5**

(i) Name the similarity criterion.

(ii) Find AB.

(iii) Find the width PB of the river.

**OR**

D, E and F are the mid-points of sides BC, CA and AB of $\triangle ABC$. Prove that $\triangle DEF \sim \triangle ABC$. If the perimeter of $\triangle ABC$ is 24 cm, find that of $\triangle DEF$. **5**

**20.** **Case study:** A circle with centre O is inscribed in quadrilateral ABCD with $\angle D = 90^\circ$. It touches BC at Q, CD at R and DA at S. $BC = 38$ cm, $CD = 25$ cm and $BP = 27$ cm (P on AB). **5**

(i) Find BQ.

(ii) Find CQ and CR.

(iii) Find the radius, giving reasons.

**OR**

Prove that the lengths of tangents drawn from an external point to a circle are equal. Hence show that if PA and PB are tangents and AB the chord of contact, $\angle PAB = \angle PBA$. **5**

**END OF QUESTION PAPER** — **Total: 40 Marks**